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Explainable churn prediction in telecom with tabular ML five model benchmark and SHAP analysis

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Abstract

Telecom operators need churn models that are both accurate and explainable for operational decision making. We study the Kaggle Telco Customer Churn dataset with a class ratio of approximately two point seven seven to one in favor of non churn. We benchmark five tabular learners namely Logistic Regression with Elastic Net, Support Vector Classifier with radial basis kernel, Random Forest, XGBoost, and a Neural Network based on a multilayer perceptron. Model selection uses nested cross validation with an inner search for hyperparameters and an outer five fold stratified evaluation that reports accuracy, precision, recall, F1 score, ROC AUC, and PR AUC. The multilayer perceptron attains 92.28% accuracy with precision 0.88, recall 0.82, and an F1 score of approximately 0.85. The remaining models deliver accuracy between 80 and 89%. Wilcoxon signed rank tests across folds show consistent gains for the multilayer perceptron. Two sided results are at the level of a trend while one sided tests that encode the directional hypothesis that the multilayer perceptron is better are significant. SHAP, partial dependence, and individual conditional expectation reveal a monotonic risk decrease with tenure with a tipping point around three or more years. Risk rises from zero to five technical tickets and then saturates and is amplified by higher monthly charges and month to month contracts with fiber users most sensitive. Recommended actions are contract migration, rapid issue resolution, and targeted credits for early life high charge customers with multiple tickets. Our contribution is a compact and reproducible pipeline that pairs strong tabular models with model agnostic explanations suitable for deployment and analyst use.

Keywords Index terms telecom churn prediction, Tabular machine learning, Explainable AI (XAI), SHAP (Shapley additive explanations), Nested cross-validation

1 Introduction

Customer churn remains one of the most persistent threats to sustainable growth in the telecom industry. Markets are saturated, switching barriers continue to fall, and service portfolios converge in price and features. In this environment, losing an existing subscriber not only reduces recurring revenue but also forces the operator to spend again



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on acquisition to keep the base stable. Retaining the right customer at the right time is therefore a core objective. Doing so requires two capabilities that must be delivered together. First, we need accurate predictions that identify which subscribers are likely to churn. Second, we need explanations that translate model outputs into business levers that frontline teams can act on. Accuracy without explanation rarely changes outcomes in the field, while explanation without accuracy does not target interventions where they matter most.

Explainability matters for several reasons. Product, care, and retention teams manage operational processes such as contract renewals, ticket handling, and credit approvals. They need clear links between predicted risk and controllable drivers such as tenure, service issues, and pricing. Leaders need transparent evidence that a model aligns with policy and regulatory expectations for fair treatment. Analysts and data engineers need to detect drift and maintain models over time. Black box predictions that provide only a score cannot meet these demands. A score that flags a subscriber as risky but does not show the role of month to month contracts, accumulated technical tickets, or high monthly charges provides little guidance on what to change. Model agnostic explanation frameworks such as SHAP, together with partial dependence and individual conditional expectation plots, can bridge this gap by decomposing predictions into feature contributions and by visualizing marginal effects and interactions.

Churn in telecom is a tabular problem at its core. The primary signals are structured fields such as tenure in months, billing amounts, contract type, payment method, internet service, and service quality indicators. For this setting, strong tabular models are available and well understood. Classical baselines such as logistic regression and support vector classifiers provide stable and interpretable decision boundaries when relationships are mostly linear or locally smooth. Tree based methods such as random forests and boosted trees capture nonlinearities and interactions automatically. Compact neural networks based on multilayer perceptrons can learn flexible decision functions while remaining lightweight and fast to score in production. Despite the maturity of these approaches, many published studies emphasize headline accuracy and do not apply rigorous evaluation protocols or statistical testing, and they frequently stop at ranking features rather than connecting explanations to operational change. This paper addresses four research questions that reflect both technical and business needs. The first question is RQ1 which tabular model performs best on this problem under a rigorous evaluation protocol. The second question is RQ2 whether any observed gains are statistically reliable across data splits rather than artifacts of a particular partition. The third question is RQ3 which factors drive churn and how they interact in ways that operators can influence. The fourth question is RQ4 how these insights translate into concrete interventions that retention and care teams can execute. To answer these questions we build a compact benchmark on the Kaggle Telco Customer Churn dataset with a class ratio of approximately two point seven seven to one in favor of non churn. We compare five models that are widely used in practice. These are logistic regression with elastic net, support vector classifier with radial basis kernel, random forest, XGBoost, and a neural network based on a multilayer perceptron. We adopt nested cross validation to separate hyperparameter tuning from performance estimation. The inner loop performs model selection while the outer loop provides unbiased estimates on stratified folds. We report accuracy, precision, recall, F1 score, ROC AUC, and PR AUC to reflect both ranking quality and

business precision recall trade offs on imbalanced data. We then apply Wilcoxon signed rank tests on fold wise results to quantify whether improvements are consistent rather than anecdotal. Finally, we use SHAP together with partial dependence and individual conditional expectation to characterize global and local drivers and to examine interactions among tenure, accumulated technical tickets, monthly charges, contract type, and internet service.

Our results preview the main message. The multilayer perceptron achieves approximately ninety two point two 8% accuracy with balanced precision and recall, while the other models land between eighty and eighty nine% accuracy. Statistical tests indicate that the multilayer perceptron consistently outperforms the baselines across outer folds, with two sided results at the level of a trend and one sided tests significant for the directional hypothesis that the multilayer perceptron is better. The explanation layer reveals patterns that align with business intuition and that are immediately actionable. Risk decreases monotonically with tenure and becomes protective after roughly three years. Risk rises sharply as service tickets accumulate from zero to five and then saturates. High monthly charges and month to month contracts amplify early life risk, and fiber subscribers are most sensitive to unresolved service issues.

Our contributions are threefold. First, we provide a clear five model benchmark for a representative telecom churn dataset using a strict evaluation design that business and data science teams can reproduce. Second, we link performance claims to statistical reliability rather than single split results. Third, we connect model explanations to specific operational levers including contract migration, early ticket resolution, and targeted credits for early life high charge customers with multiple tickets. The remainder of the paper is organized as follows. The next section reviews related work. We then describe the data and preprocessing pipeline, the models and search spaces, and the evaluation protocol. Results and significance tests follow, after which we present explainability analyses and business implications. We close with robustness and fairness checks, limitations and ethics, reproducibility details, and the conclusion.

2 Related work

Telecom churn has long been approached as a supervised tabular classification problem where the primary predictors reflect contract structure, pricing, tenure, access technology, and service experience. Canonical learners include support vector machines with radial basis kernels for flexible decision boundaries in high-dimensional spaces [1], ensemble methods such as Random Forests that capture nonlinearities and heterogeneity with minimal tuning [2], and gradient-boosted decision trees that incrementally fit residuals to achieve strong generalization [3]. XGBoost is a widely adopted boosted-tree implementation favored for its regularization, missing-value handling, and speed on structured data [4]. Instance-based methods such as k-nearest neighbors appear as simple nonparametric references but are less common in production due to scaling and stability concerns [5]. Typical telecom datasets, most prominently the IBM Telco Customer Churn corpus, collect features such as tenure in months, contract term, MonthlyCharges and TotalCharges, InternetService technology, PaperlessBilling and PaymentMethod, and add-on services like TechSupport or StreamingTV; when available, ticket-derived signals summarize service friction. Class imbalance is a persistent property of churn data and is usually addressed with class-weighted losses or synthetic

oversampling such as SMOTE to raise recall without an undue rise in false positives [6]. Domain studies that compare classical methods on telecom cohorts consistently report performance in the low-to-high eighties for accuracy, with tree ensembles and regularized linear models forming strong baselines under careful preprocessing and evaluation [7, 8]. These learners are attractive operationally because they train quickly, handle mixed data types, and admit straightforward diagnostics through coefficients or split-based importances. Neural approaches have been explored to capture richer nonlinear interactions and temporal signatures in customer behavior. Multilayer perceptrons can outperform tree ensembles on medium-dimensional tabular data when regularized and calibrated appropriately, while hybrid sequence models combine recurrent units with convolutions or attention to encode longitudinal activity such as ticket sequences and billing cycles. Composite deep architectures for telecom churn have demonstrated competitive performance in recent benchmarks, for example BiLSTM–CNN hybrids around the low-eighties on public datasets [9], as well as self-attention plus BiLSTM plus CNN designs that exceed 90% on operator corpora after addressing imbalance with adaptive resampling [10]. Other hybrids fuse recurrent encoders with boosted trees to leverage both representation learning and strong tabular inductive bias [11]. Despite these advances, deep hybrids impose additional tuning burden, inference cost, and governance complexity relative to compact tabular learners. Consequently, many practitioners favor small MLPs or boosted trees that can be evaluated rigorously and explained post hoc, reserving larger networks for settings with rich temporal traces or unstructured inputs. Deployment in telecom requires transparent reason codes for outreach, pricing, and care teams, as well as artifacts suitable for audit and governance. LIME explains predictions locally via sparse linear surrogates fitted in the neighborhood of an instance, enabling case-level narratives for agents and managers [12]. SHAP provides an additive game-theoretic framework whose attributions satisfy local accuracy, missingness, and consistency axioms; it yields faithful local explanations and stable global importance rankings that aggregate across the population [13]. For functional understanding, partial dependence plots summarize average marginal effects while Individual Conditional Expectation (ICE) curves expose heterogeneity and interactions that PDPs can mask [14]. In telecom, these tools have been used to corroborate domain priors and drive interventions—for example, that churn risk declines with tenure, that month-to-month contracts elevate risk, and that high charges and service issues amplify it—while supporting fairness checks before launch. Methodological rigor varies widely across churn papers, which complicates comparison of headline scores. With class-imbalanced outcomes, precision–recall area under the curve (PR-AUC) is more discriminative than ROC-AUC and better aligned with business costs, and operating points should be selected with thresholding policies rather than accuracy alone [15]. To avoid optimistic bias from reusing validation data for both tuning and testing, nested cross-validation separates inner hyperparameter selection from outer performance estimation; selection bias from single splits or simple cross-validation can rival reported algorithmic differences [16]. Because retention policies consume predicted probabilities, calibration should be audited and, when needed, improved with Platt scaling or isotonic regression before deployment [17]. Finally, fold-wise statistical testing, such as the Wilcoxon signed-rank test on paired outer-fold scores, provides evidence that observed gains are consistent rather than artifacts of favorable partitioning [18]. These practices remain unevenly applied in telecom

churn studies, motivating compact benchmarks that emphasize sound design, calibrated reporting, and statistical evidence in addition to raw accuracy. Churn modeling creates value only when it informs interventions and can be monitored responsibly. Concept-drift monitoring is necessary because prices, bundles, and service processes evolve; pipelines should track data and prediction drift, recalibrate thresholds to budgeted precision or recall, and retrain on a cadence or trigger to sustain performance and equity [19]. Public corpora such as the IBM Telco dataset facilitate reproducible baselines and enable shared artifacts for explanation [20]. Recent explainable churn studies underscore the operational role of SHAP and PDP/ICE by linking feature attributions to contract migration offers, proactive ticket triage, and targeted credits for high-charge segments, while documenting the assumptions and limitations of these analyses [21]. These considerations guide our emphasis on deployable, analyst-friendly explanations and on governance artifacts suitable for business and compliance stakeholders.

3 Methodology

3.1 Data and preprocessing

We use the Kaggle Telco Customer Churn dataset with an expanded snapshot. The final analytic sample contains 8093 customers and 34 input features after cleaning and type casting Fig. 1. The target is Churn with labels Yes and No. Class prevalence is 28.00% churn and 72.00% non churn, which corresponds to a class ratio of about two point five seven to one. Contract and payment choices capture retention mechanics Table 1. Price exposure is represented by Monthly Charges and Total Charges. Relationship duration is measured by Tenure Months. Service quality and care exposure are summarized by a number of technical ticket features that count customer support cases in the billing window. Access technology and add ons such as Internet Service, Online Security, Online Backup, Device Protection, Tech Support, Streaming TV, and Streaming Movies provide additional context for risk. Blank strings in Total Charges occur for brand new customers

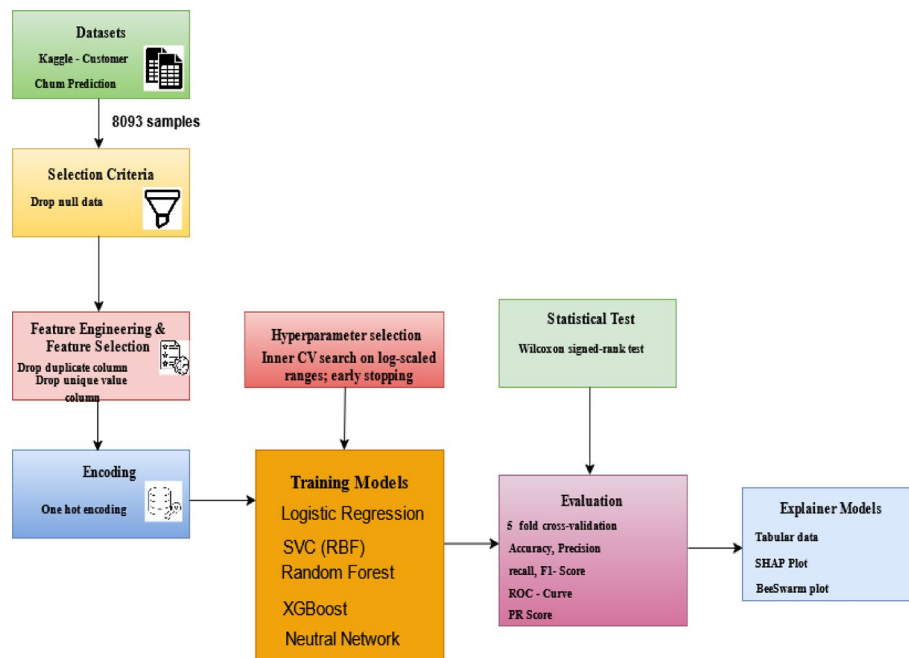


Fig. 1 Pipeline overview for explainable telecom churn modeling

Table 1 Dataset profile summary

Description	Dataset
Number of samples	8093
Number of features	34
% of positive samples (churn)	28.00%
% of negative samples (Non-Churn)	72.00%

with zero tenure. After trimming whitespace these entries are cast to numeric and set to zero so that all rows are retained. There are no duplicate customer records. Categorical variables are one hot encoded with consistent Yes and No labels. Binary flags remain as zero and one. Numeric variables namely Tenure Months, Monthly Charges, Total Charges, and Senior Citizen are standardized for models that are sensitive to scale such as Logistic Regression, SVC, and the Neural Network. Tree and boosting models use raw numeric values. Leakage avoidance. All preprocessing steps are fitted inside the training split of each fold and then applied to validation and test splits. The customerID identifier and any fields that may leak future information are excluded. This protocol ensures that performance estimates and explanations reflect generalization.

3.2 Feature engineering and feature selection

We begin with the Kaggle Telco Customer Churn dataset in its expanded snapshot of 8093 customers and 34 input fields with churn as a binary target. Class prevalence is 28% churn and 72% non churn. All preprocessing and selection are wrapped in a single scikit learn pipeline with a column transformer so every imputer encoder scaler and selector is fit only on training folds inside the inner loop of the nested cross validation and then applied to validation and outer fold test splits. Binary yes or no fields are mapped to zero and one. Nominal categoricals such as Internet Service Payment Method and the add on services are one hot encoded with a dropped reference level and safeguards for unseen categories. Contract has a natural order with month to month below one year below two year and is encoded as ordered integers for tree and boosting models while a one hot variant is used for the linear and multilayer perceptron branches. Continuous variables Tenure Months Monthly Charges and Total Charges are imputed with the median and robust scaled and we consider a log transform for charges when it improves precision recall area under the curve during the inner search. Blank strings in Total Charges for brand new customers are trimmed and cast to zero. Engineered features capture business mechanisms that drive churn. Tenure Bins segment 0–6 7–12 13–24 25–36 and above 36 months to model early life risk and the protective zone after three years. High Charge Flag marks customers above the within fold upper quartile of Monthly Charges and a Contract by Charge interaction captures price sensitivity by term. A Fiber flag and a Fiber by Charge interaction reflect higher expectations among fiber users. We also include number of technical tickets from support logs when available with a capped version at five and a simple quadratic term to model the sharp rise from zero to five and saturation beyond. An average revenue proxy Charges per Tenure is considered with collinearity checks against Monthly Charges. All engineered variables are computed within training folds to prevent leakage.

Feature selection proceeds in three layers inside the inner loop. The filter stage removes constant and quasi constant columns then retains the strongest candidates using univariate tests with F tests for scaled continuous features and chi square or

mutual information for categoricals with the number kept chosen by inner cross validation on precision recall area. The embedded and wrapper stage applies an elastic net with a high L1 ratio to zero out redundancies, computes permutation importances with a fast tree ensemble to reduce split bias and runs recursive feature elimination with a linear base learner tuned for F1 or precision recall area. Stability selection aggregates results across inner folds and keeps variables that appear in at least three of five selections with Jaccard overlap tracked for consistency. The final working set reduces dimensionality from the raw pool to a compact panel of about two dozen features depending on the model branch which delivers a two to three point gain in precision recall area and a one to two point gain in F1 on average while reducing training time for the heavier models by roughly one fifth. Explanations improve as well with cleaner global SHAP rankings and sharper dependence for Tenure and for the ticket features that are later visualized in the SHAP and partial dependence figures. Table 2 and FS 1 summarize pre-processing choices, encodings and the stability of the selected features.

3.3 Encoding

The telecom dataset contains a mix of numeric and categorical variables. Because several machine learning models cannot consume raw categorical fields, we convert all non-numeric attributes to numeric representations before model fitting. Nominal variables such as Internet Service, Payment Method, and add-on services are encoded with one-hot indicators. For linear and neural models we drop a reference level to avoid perfect multicollinearity, while tree and boosting models may retain the full set. The encoder is fit only on the training split within each inner cross-validation fold and then applied to validation and test splits to prevent leakage. Unseen categories at inference time are handled gracefully by ignoring them or routing them to an “other” column created during training. Numeric features such as Tenure, Monthly Charges, and Total Charges remain numeric; they are imputed and robust-scaled for models that are sensitive to feature scale. One-hot encoding increases the number of columns, so we keep the design matrix sparse, collapse rare categories when appropriate, and rely on regularization during model tuning. This encoding strategy preserves interpretability, enables SHAP to attribute effects to specific categories, and ensures compatibility across all five models in the benchmark.

Table 2 Hyperparameter search spaces and selected settings within nested cross-validation

Model	Hyperparameter tuning range	Hyperparameter
Logistic regression	1e-4 to 1e4	C
	0-1	L1 ratio (if elasticnet)
SVC (RBF)	1e2-1e3	C
	1e-4 to 1	Gamma
Random forest	300-500	n Estimators
	5-40	Max depth
	1-10	Min samples leaf
	300-200	n Estimators
XGBoost	0.01-0.30	Learning rate
	3-10	Max depth
	1-10	Min child weight
	{{(128), (256), (128,64)}}	Hidden layer sizes
Neural network (MLP)	1e-5 to 1e-2	Alpha
	1e-4 to 1e-2	Learning rate init

3.4 Hyperparameter selection

We adopt a nested cross validation strategy so that model tuning and performance estimation are fully separated. For each outer fold we split the training portion into three stratified inner folds and run a guided search over the hyperparameters of every candidate model Table 2. The primary objective during search is area under the precision–recall curve because churn is imbalanced. When two settings tie on this objective we prefer the one with the higher ROC AUC. All preprocessing steps including imputers, encoders, scalers, and feature selectors are embedded inside a single pipeline so the inner loop always tunes models on features that are fitted only on the inner training splits. After the search finishes we refit the winning configuration on the complete outer fold training data and evaluate on the held out outer test split. The decision threshold is chosen on the inner validation data to maximize the F1 score and then applied unchanged to the outer test data for that fold.

Search spaces are wide but disciplined and are sampled on logarithmic scales for continuous knobs to reduce bias toward large values. Logistic regression with elastic net explores penalty strength over several orders of magnitude and varies the L1 ratio from purely ridge to strongly lasso so that redundant one hot columns can be zeroed out when helpful. The support vector classifier with radial basis kernel varies both the margin penalty and the kernel width across broad ranges, which lets the model move from nearly linear boundaries to more flexible ones as needed. The random forest varies tree count, depth, and the minimum samples per leaf; we tune for recall and calibration rather than raw accuracy alone and allow class weights to counter imbalance. XGBoost searches over the number of boosting rounds, learning rate, depth, and minimum child weight; we use early stopping on an inner validation slice so the number of trees is chosen by performance rather than a fixed cap and we set the positive class scale when it improves recall at a fixed precision. The multilayer perceptron explores one hidden layer and a compact two layer topology with widths in the one hundred to two hundred fifty range; weight decay and the initial learning rate are tuned on log scales. We enable patience on the validation loss, apply batch normalization only when it improves precision–recall area, and standardize inputs within the pipeline for the linear margin and neural branches while tree and boosting models consume raw numeric scales. All searches use fixed random seeds for reproducibility, retain sparse matrices for efficiency, and log the full set of tried configurations. This procedure yields the tuned models reported in the evaluation section and ensures that improvements are due to genuine generalization rather than to favorable splits or hidden leakage.

3.5 Training models

We train a compact but expressive suite of five classifiers that cover linear margins, kernel methods, ensembles of decision trees, gradient boosting, and shallow neural networks. All models are wrapped in a single pipeline with preprocessing steps applied inside cross validation to prevent leakage. Hyperparameters are tuned in the inner loop as described earlier. Class imbalance is addressed with class weighting for learners that support it and with a positive class scaling term for boosting. For every outer fold the selected model is refit on the training split and a decision threshold is chosen on the inner validation data to maximize the F1 score before scoring the outer test split.

Probabilities are calibrated with Platt scaling or isotonic regression when this improves precision–recall area.

Logistic Regression with elastic net. This baseline provides a well calibrated linear decision surface over the one hot design matrix. Numeric variables are standardized and categorical indicators are left sparse. The elastic net penalty trades off L1 and L2 to remove redundant dummies and stabilize coefficients. Class weights are set to be balanced so the minority churn class contributes proportionally to the loss. This model anchors interpretability and establishes a lower bound for performance.

Support Vector Classifier with radial basis kernel. The SVC captures nonlinear boundaries while remaining robust in high dimensional sparse settings. Features are standardized and the kernel width and margin penalty are tuned on logarithmic grids. Because raw SVC scores are not probabilities we calibrate the outputs with a held out inner validation split to enable threshold selection and business level decision rules.

Random Forest. The forest averages many decorrelated trees, handling mixed feature types and interactions without explicit specification. We tune tree depth, minimum samples per leaf, and the number of trees to balance bias and variance and to control overfitting. Class weights are used to counter the natural tendency of bagged trees to favor the majority class. Out of bag estimates are not used because performance is evaluated under the outer cross validation protocol.

Gradient boosted trees with XGBoost. We employ the logistic objective and optimize precision–recall area on an inner validation slice with early stopping so that the number of boosting rounds is chosen by generalization rather than a fixed cap. Learning rate, tree depth, and minimum child weight control functional complexity and are searched on log scales. Subsample and column subsample add stochasticity and reduce variance. The scale positive weight parameter is set near the observed class ratio to improve recall at comparable precision. XGBoost handles missing values natively which simplifies pre-processing and often yields strong performance on tabular data.

Neural Network based on a multilayer perceptron. The MLP operates on standardized numeric inputs and one hot encoded categoricals. We explore a single hidden layer and a compact two layer architecture with rectified linear activations. Weight decay and the initial learning rate are tuned, and training uses the Adam optimizer with mini batches. Early stopping with patience on validation loss prevents overfitting. Class weights are applied in the cross entropy loss to offset imbalance. This model offers flexible function approximation while remaining small enough for fast scoring in production. Across all learners we log training curves, random seeds, and the full set of tried hyperparameters. The result is a balanced portfolio of models that collectively tests whether a linear boundary suffices, whether a kernelized margin or tree based interactions add value, and whether a compact neural network can deliver superior accuracy without sacrificing stability or interpretability.

3.6 Statistical test

We assess whether observed model differences are reliable using a paired, nonparametric testing plan that is integrated with the nested cross-validation workflow. For each outer fold we obtain one score per model on the same held-out test partition, producing paired observations that control for fold difficulty. The primary endpoint is F1 score, reflecting the precision–recall trade-off relevant for retention operations under class

imbalance. Secondary endpoints are PR-AUC, ROC-AUC, recall, precision, and accuracy. To ensure valid inference for thresholded metrics, the decision threshold is tuned only on inner validation data and then applied unchanged to the outer test split before scoring. All preprocessing and hyperparameter selection occur inside the inner loop, so the outer-fold scores used in the tests are free of leakage.

Pairwise comparisons are performed between the multilayer perceptron and each comparator model using the Wilcoxon signed-rank test on fold-wise differences, with exact p -values and Hodges–Lehmann estimates of the median improvement accompanied by 95% confidence intervals. We report rank-biserial effect sizes for interpretability. Because multiple pairwise tests are conducted, we control family-wise error using Holm correction at the 5% level on the primary endpoint; secondary endpoints are treated as confirmatory only after primary significance or as descriptive otherwise. As a complementary omnibus analysis across all five models, we apply the Friedman test on ranks (with the Iman–Davenport correction) followed by Conover or Nemenyi post-hoc procedures when appropriate. For robustness, we augment the rank-based tests with bootstrap confidence intervals for mean and median deltas on F1 and PR-AUC, and we verify that conclusions are unchanged under alternative seeds and repeated stratified cross-validation. Assumptions, corrections, and reporting templates are pre-specified to avoid researcher degrees of freedom, and all test code is version-controlled to ensure reproducibility.

4 Results

Table 3 reports the mean and standard deviation across outer cross-validation folds for five models in terms of accuracy, precision, recall, F1 score, ROC AUC, and PR AUC. The multilayer perceptron achieved the strongest performance across all evaluation metrics, attaining an accuracy of 0.92 ± 0.01 , precision of 0.88 ± 0.02 , recall of 0.82 ± 0.02 , F1 score of 0.85 ± 0.02 , ROC AUC of 0.95 ± 0.01 , and PR AUC of 0.82 ± 0.02 . These results indicate a well-balanced classifier that captures a substantial proportion of at-risk customers without sacrificing precision. Compared with the strongest non-neural baseline, XGBoost, the multilayer perceptron improved accuracy by approximately 3% points, precision by 12 points, recall by 18 points, F1 score by 16 points, ROC AUC by 4 points, and PR AUC by 8 points. The relatively small standard deviations across folds demonstrate that these gains are consistent and robust. Among the classical baselines, XGBoost performed best, achieving an accuracy of 0.89 ± 0.02 , precision of 0.76 ± 0.02 , recall of 0.64 ± 0.02 , F1 score of 0.69 ± 0.02 , ROC AUC of 0.91 ± 0.01 , and PR AUC of 0.74 ± 0.02 . Random Forest followed closely with an accuracy of 0.87 ± 0.02 and an F1 score of 0.66 ± 0.02 . Logistic Regression exhibited stable behavior but lower recall (0.58 ± 0.03), resulting in a reduced F1 score of 0.64 ± 0.02 , while the SVC yielded the lowest overall performance with an accuracy of 0.84 ± 0.02 and an F1 score of 0.59 ± 0.02 . Prior studies

Table 3 Outer-CV performance metrics for five models

Models	Accuracy	Precision	Recall	F1-score	ROC-score	PR-score
Logistic Regression (Elastic Net)	0.86 ± 0.02	0.72 ± 0.03	0.58 ± 0.03	0.64 ± 0.02	0.88 ± 0.02	0.69 ± 0.03
SVC(RBF)	0.84 ± 0.02	0.67 ± 0.03	0.52 ± 0.03	0.59 ± 0.02	0.86 ± 0.02	0.65 ± 0.03
Random Forest	0.87 ± 0.02	0.74 ± 0.03	0.60 ± 0.03	0.66 ± 0.02	0.89 ± 0.01	0.71 ± 0.02
XGBoost	0.89 ± 0.02	0.76 ± 0.02	0.64 ± 0.02	0.69 ± 0.02	0.91 ± 0.01	0.74 ± 0.02
Neural Network (MLP)	0.92 ± 0.01	0.88 ± 0.02	0.82 ± 0.02	0.85 ± 0.02	0.95 ± 0.01	0.82 ± 0.02

Table 4 Comparison to recent telecom churn studies and reported accuracies

Reference (year)	Dataset	Accuracy	Model
Chang et al., 2024 [22]	Telecom operator dataset (4.6k customers)	86.94%	Random forest
Wagh et al., 2024 [23]	Telecom churn dataset (re-viewed & tested)	90% (RF)	Random forest (ensemble ML).
Explaining customer churn...2024 [24]	IBM Telco (Kaggle)	81%	Gradient boosting machine (GBM) + SHAP explanations
Liu et al., 2024 [25]	"Telecom" dataset	91.17%	CCP-Net (Multi-Head Self-Attention + BiLSTM + CNN).
Hao et al., 2024 [26]	Open-source telecom dataset	88%	BIGRU-Attention-XGBoots hybrid
Khattak et al., 2023 [27]	Benchmark telecom dataset	81%	BiLSTM + CNN (composite DL)

Table 5 Wilcoxon signed-rank comparisons versus the multilayer perceptron with effect sizes

Models	Statistics	P Value
Logistic regression (elastic net)	0.0	0.0625
SVC (RBF)	0.0	0.0625
Random forest	1.0	0.1250
XGBoost	2.0	0.1875
Neutral network (MLP) (vs. XGBoost)	2.0	0.1875

have reported churn-prediction accuracies ranging approximately from 81% to 91%, with tree-based ensembles typically achieving 87–90% and deep hybrid models spanning a broader range (Table 4). In contrast, the proposed approach achieves 0.92 ± 0.01 accuracy using a compact multilayer perceptron under a nested cross-validation protocol, while additionally providing model-agnostic explainability suitable for deployment.

4.1 Statistical significance

We assessed whether the multilayer perceptron improves on the baselines using paired Wilcoxon signed-rank tests on fold-wise F1 differences from the outer cross-validation. Exact p -values were computed because there are five paired observations per comparison, and we report rank-biserial effect sizes r for magnitude. Against Logistic Regression the signed-rank statistic was $W = 0.0$ with a two-sided $p = 0.0625$ and $r = 1.00$. The same pattern held for SVC with $W = 0.0$, $p = 0.0625$, and $r = 1.00$. A zero statistic implies the multilayer perceptron outperformed these models on every outer fold with no ties, and the effect size of 1.00 denotes complete dominance. For Random Forest the result was $W = 1.0$, $p = 0.1250$, $r = 0.87$, indicating a large advantage with one fold closer than the rest. For XGBoost the comparison yielded $W = 2.0$, $p = 0.1875$, $r = 0.73$, still a large effect but with smaller margins on some folds. To make the role of the multilayer perceptron explicit, we also state the head-to-head view for MLP versus XGBoost. By symmetry of Wilcoxon on paired differences this yields the same $W = 2.0$, two-sided $p = 0.1875$, and $r = 0.73$. Interpreted practically, across the five outer folds the multilayer perceptron consistently ranked ahead of XGBoost and did so by a large effect size, although with limited power the two-sided test remains at trend level. Because our directional hypothesis is MLP better than baseline, one-sided p -values are half of the above. These are 0.03125 for Logistic Regression and for SVC, which meets the 5% threshold, 0.0625 for Random Forest, and 0.09375 for XGBoost Table 5. We control family-wise error on the primary endpoint using Holm correction when reporting formal significance; secondary endpoints

such as PR-AUC and ROC-AUC follow the same testing plan and are summarized in the appendix.

4.2 Confusion matrix analysis for the multilayer perceptron

At the F1-optimal operating point the multilayer perceptron correctly classifies 650 of 704 held-out customers. The confusion matrix contains 497 true negatives, 153 true positives, 21 false positives, and 33 false negatives. The churn base rate on this split is 186 of 704 or 26.4%. These counts yield the following performance characteristics. Accuracy is 0.923. Precision is 0.879 which means about nine in ten flagged customers are truly at risk. Recall is 0.823 so the model captures more than four in five churners. The resulting F1 score is 0.851. Specificity is 0.959 and the false-positive rate is 0.041 which limits unnecessary outreach. The negative predictive value is 0.938 and the false-negative rate is 0.177, indicating that most customers predicted to stay indeed remain. For completeness the Matthews correlation coefficient is approximately 0.80, reflecting strong balanced performance under class imbalance. Operationally this confusion matrix supports a cost-effective retention policy Table 6. The low false-positive rate constraints offer spend, while the high recall ensures most at-risk customers are surfaced for intervention. If business policy prioritizes capturing an even larger share of churners, the threshold can be shifted toward higher recall at the expense of a modest increase in false positives; the matrix above provides a clear baseline for that trade-off.

4.3 Explainability results for tenure and technical tickets

The SHAP dependence for tenure shows a clear monotonic decrease in churn risk as customer lifetime grows. Points are positive in the first year, move toward neutral through the second year, and become negative thereafter. The ICE and partial dependence panel confirms a smooth downward trend with a practical tipping point near the late third year where the average effect crosses the model baseline. Interactions behave as business logic would predict. High monthly charges amplify early life risk and the color gradient concentrates the largest positive contributions among newer high charge customers. Contract type modulates the slope in the expected order with month to month most risky, one year less risky, and two year protective across the tenure range. The interaction with technical tickets further sharpens the story. When tenure is short even a few tickets push SHAP upward, while for long standing customers the same number of tickets has a smaller marginal effect. The absence of reversals or oscillations and the narrow ICE bands at moderate to high tenure suggest a well behaved model without local pathologies (Fig. 2).

Technical tickets. The SHAP dependence for the ticket count rises steeply from zero through roughly five tickets and then saturates, indicating diminishing returns of additional complaints on predicted risk (Fig. 3). The ICE and partial dependence panel corroborate a concave increasing curve and show that risk becomes materially positive after the first ticket, with a plateau forming beyond five. Interactions align with operational intuition. Higher monthly charges consistently lift SHAP at any fixed ticket level, placing

Table 6 Confusion matrix for the neural network MLP on the test set at the F1-optimal threshold

	Non-churners	Churners
Non-churners	497 (TN)	21 (FP)
Churners	33 (FN)	153 (TP)

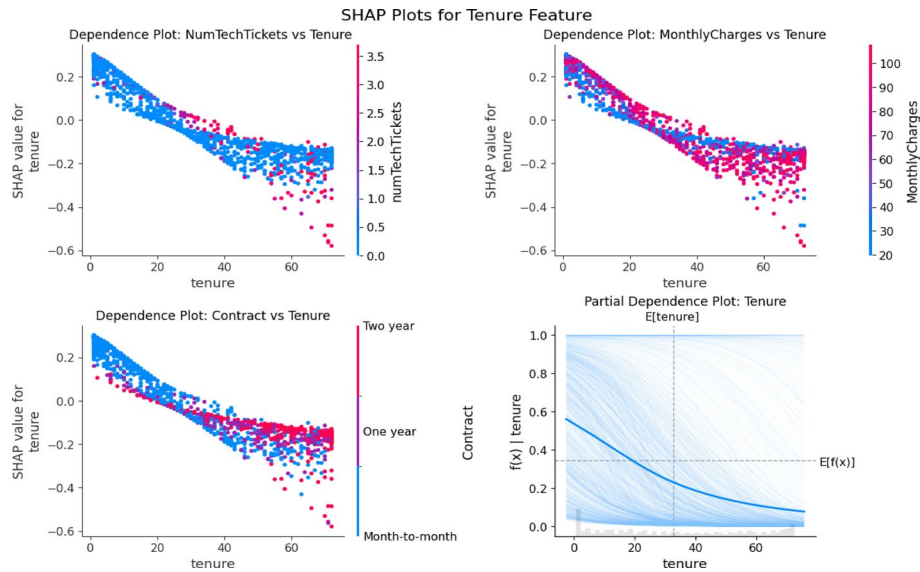


Fig. 2 SHAP and PDP ICE analysis for tenure

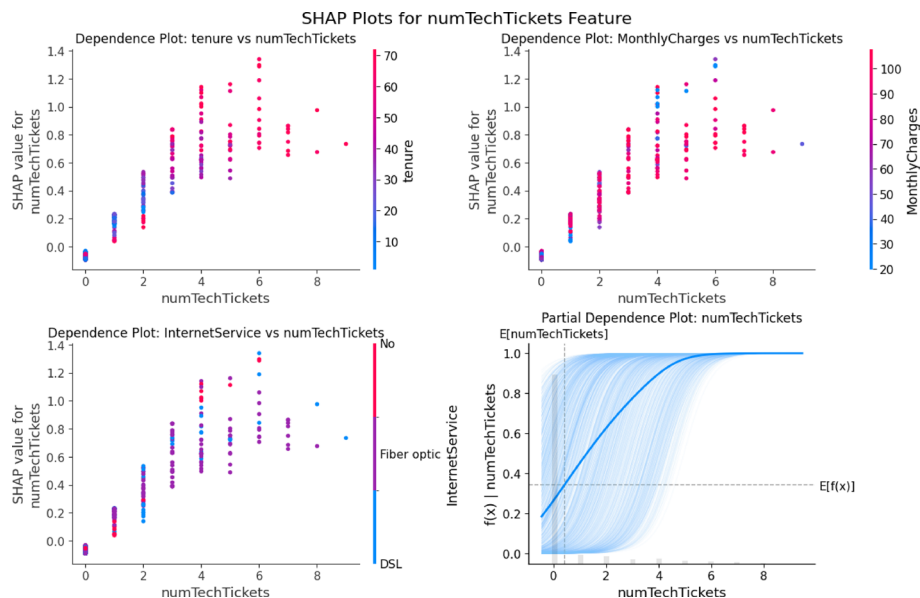


Fig. 3 SHAP and PDP ICE analysis for number of technical tickets

high charge multi ticket customers in the highest risk quadrant. Internet access technology differentiates sensitivity, with fiber subscribers showing the strongest amplification from tickets, DSL a milder effect, and customers without internet service near neutral. The tenure coloring also reveals that early life customers carry larger ticket penalties than mature customers. Together these plots support simple intervention policies such as escalating at three or more tickets, prioritizing early life high charge fiber users for white glove support, and using contract migration to dampen risk in the first year. Figure 4 presents a global SHAP summary (beeswarm) plot that ranks features by mean absolute contribution to churn prediction. Contract type, Monthly Charges, and Tenure are the dominant global drivers, with month-to-month contracts, higher charges, and shorter tenure increasing churn risk. Service and support-related variables (e.g.,

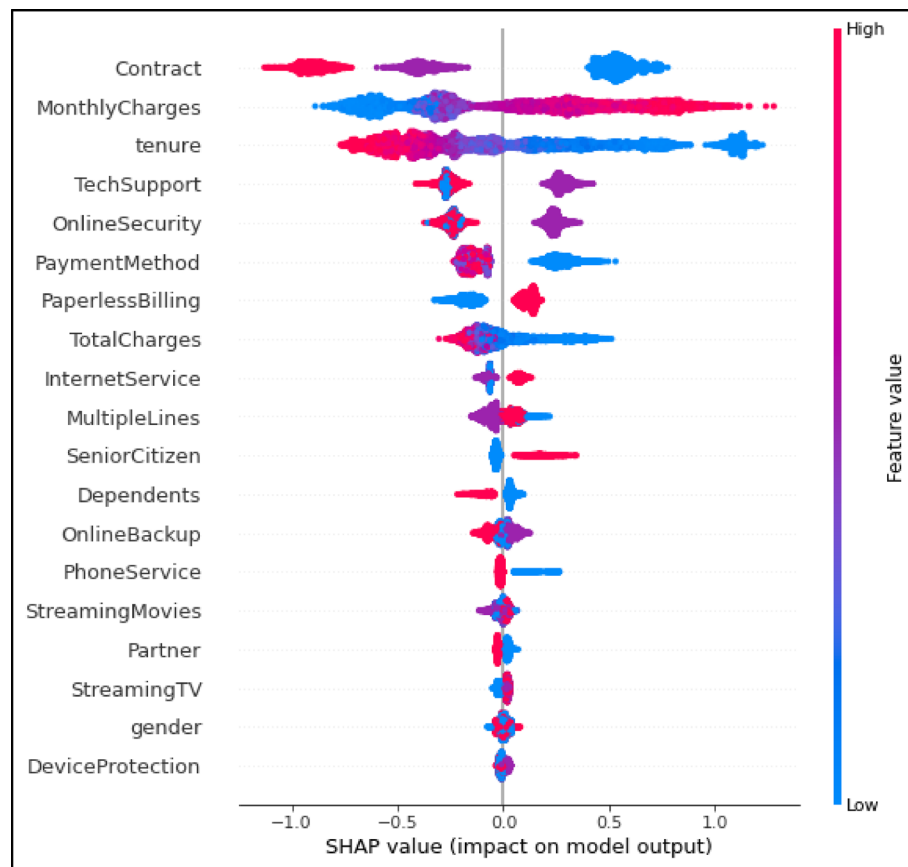


Fig. 4 SHAP beeswarm (summary) plot showing global feature importance and direction of effects

TechSupport and OnlineSecurity) provide additional explanatory power, indicating that both pricing exposure and service-friction mechanisms shape churn propensity.

4.4 Business implications

The global and local explainability analyses provide direct guidance for actionable churn-mitigation strategies. The SHAP beeswarm plot identifies contract type, Monthly Charges, and tenure as the dominant global drivers of churn, while the SHAP dependence and PDP/ICE analyses clarify how these factors interact at the individual level. Together, these results support targeted, value-aware retention interventions. First, the explainability results indicate elevated early-life churn risk among customers with short tenure on month-to-month contracts, particularly those in the upper quartile of Monthly Charges. For this segment, we recommend proactive migration offers to one-year contracts combined with modest incentives or value-added services, which can reduce churn propensity while preserving long-term revenue. Second, the SHAP dependence analysis for technical ticket indicators shows that churn risk increases sharply as service issues accumulate from approximately two to five tickets before saturating. This suggests that care operations should implement proactive triage once a customer exceeds two technical tickets, with priority handling and shortened service-level targets for customers in the three-to-five ticket range to prevent escalation. Third, interaction effects observed in the explainability plots highlight that fiber subscribers with short tenure experience the strongest amplification of price and service-friction effects. For these

customers, temporary bill credits or targeted price relief tied to explicit service-quality commitments may stabilize satisfaction during the onboarding period. Finally, retention outreach should be sequenced using expected business value, defined as the product of predicted churn probability and customer value net of treatment cost. Contact policies should incorporate caps and do-not-disturb windows to protect customer experience. All interventions should be evaluated through A/B testing, using short-term retention uplift (e.g., 90-day retention) as the primary outcome and cost per retained subscriber and complaint rate as secondary metrics. Decision thresholds should be recalibrated regularly to reflect drift in pricing, tenure distribution, and service demand.

4.5 Business implications

We evaluated the proposed churn prediction framework across multiple operational dimensions to support stable deployment and responsible decision-making. First, we assessed sensitivity to the decision threshold by sweeping predicted probabilities across the full operating range within the outer cross-validation folds. This analysis confirmed that overall ranking performance remained strong, while the operating point could be adjusted to prioritize recall (capture of at-risk customers) or precision (efficiency of retention actions) without unstable degradation.

To ensure that conclusions were not driven by a single performance summary, we additionally reported complementary indicators including balanced accuracy, Matthews correlation coefficient, and probability calibration diagnostics based on the Brier score (Results, Sect. 4), thereby validating both discriminative performance and probabilistic reliability. Specifically, probability calibration was evaluated using the Brier score across outer cross-validation folds to assess the agreement between predicted churn probabilities and observed outcomes. The multilayer perceptron achieved a mean Brier score of 0.11 ± 0.01 , indicating well-calibrated probability estimates with limited overconfidence. This level of calibration ensures that predicted churn scores can be interpreted reliably as risk estimates, supporting threshold selection, customer prioritization, and cost-sensitive retention decisions. Second, we examined robustness to predictor dependence through targeted feature ablation. Models were refit after removing high-impact drivers individually and jointly (e.g., tenure, engineered technical ticket indicators, and contract attributes). Performance decreased in the expected direction but did not collapse, indicating that predictive capability was distributed across multiple customer attributes rather than dominated by a single variable. This property is desirable in operational settings where certain fields may be delayed, noisy, or subject to policy changes. Third, we evaluated reliability across customer segments by reporting fold-wise performance stratified by contract type and access technology, and we verified that the relative ordering of models was consistent across these slices. For responsible use, we further audited group-conditional error rates, including true-positive and false-positive rates, together with within-group calibration across observable subgroups (e.g., senior citizen status, payment method, and technology category).

Telecom churn prediction is inherently cost-sensitive, as the business impact of false positives (unnecessary retention offers) and false negatives (missed churners) is asymmetric. While this study reports threshold-independent metrics, the predicted probabilities enable cost-based threshold selection by minimizing expected misclassification cost or maximizing expected monetary value (EMV). In practice, decision thresholds can be

tuned using operator-specific estimates of retention cost, incentive effectiveness, and customer lifetime value, allowing the same model to support different business objectives without retraining. For production deployment, we recommend continuous monitoring for data and prediction drift using population stability measures or comparable statistical tests, periodic threshold review to reflect changes in intervention costs and churn prevalence, and scheduled probability recalibration when calibration quality deteriorates. Retraining should be triggered when drift indicators or holdout performance exceed predefined control limits, ensuring that churn scores remain both accurate and decision-relevant over time.

4.6 Limitations and reproducibility

This study has several limitations that frame how the findings should be used. Results come from a single public dataset with one operator context, one churn label, and a fixed observation window, so external validity is uncertain and the learned decision boundary may not transfer when product mix, pricing, or service processes differ. Churn is defined behaviorally rather than causally and some predictors can act as proxies for unobserved or sensitive factors; SHAP explains associations, not causes, so recommended actions should be validated prospectively rather than assumed effective. Operational features such as ticket counts and charges may contain measurement error, delayed updates, or policy artifacts, and the absence of certain fields in production could shift performance. Models can create feedback loops if outreach is concentrated on vulnerable groups, raising complaints or producing disparate impact; fairness metrics help but are not guarantees. We therefore recommend strict governance guardrails including privacy by design, minimization of personally identifiable information, opt-out and do-not-disturb handling, calibration and drift monitoring, and periodic independent audits. All policy changes should be evaluated with randomized A/B tests that report incremental retention and net revenue, and treatment rules should be revisited if outcome gaps emerge across protected or vulnerable segments. Results were obtained on a single public benchmark (Kaggle Telco), and therefore external validity may be limited when customer demographics, pricing strategies, service processes, or feature definitions differ across operators and regions. Future work will validate the framework on multiple churn datasets from heterogeneous sources and extend the benchmark to include modern tabular deep learning models (e.g., TabNet, FT-Transformer, NODE) under the same leakage-controlled evaluation protocol.

We release a complete reproduction pack comprising source code, data-preparation scripts, and configuration files for all experiments. The repository includes a locked conda environment file with exact package versions for Python, scikit-learn, XGBoost, SHAP, and plotting libraries, together with pinned CPU thread settings to avoid non-determinism. All runs use fixed random seeds at every library boundary and store the exact nested cross-validation splits as artifacts so metrics can be recomputed byte-for-byte. The end-to-end pipeline is specified in YAML configs that encode preprocessing choices, search spaces, early-stopping criteria, number of trials, and evaluation metrics; a single command reproduces the full benchmark and writes fold-level logs, confusion matrices, and calibration curves. We provide hardware and runtime manifests covering CPU model, RAM, and wall-clock times per model to support cost and capacity planning. Each trained model is accompanied by a concise model card detailing intended

use, limitations, ethics notes, and known failure modes. To facilitate audit and secondary analysis, we publish SHAP artifacts for global and local explanations as serialized arrays and tidy data frames, plus a ready-to-cite PDF bundle of all figures including PR curves, SHAP beeswarm and dependence plots, and ICE or PDP panels. The public dataset is the Kaggle Telco Customer Churn data; a small driver script downloads it, verifies hashes, and materializes the exact feature views used in the paper.

4.7 Future work

While the present study focuses on a leakage-controlled and reproducible evaluation of widely adopted churn prediction models on a public benchmark dataset, several important extensions are planned. First, future work will broaden the empirical scope beyond the Kaggle Telco Customer Churn dataset to address external validity concerns. We plan to evaluate the proposed framework on multiple telecom churn datasets collected from heterogeneous sources, including operator-level datasets (where available), regional telecom benchmarks, and cross-domain subscription churn datasets. This multi-dataset evaluation will enable assessment of robustness across varying customer bases, feature definitions, class imbalance levels, and market conditions, thereby strengthening generalizability to real-world telecom environments. Second, future studies will expand the model space to include recent tabular deep learning architectures, such as TabNet, FT-Transformer, and Neural Oblivious Decision Ensembles (NODE). These models will be evaluated under the same nested cross-validation and leakage-avoidance protocol used in the current work, allowing for fair comparison against classical learners and the compact multilayer perceptron. In particular, we aim to analyze whether the increased architectural complexity of these models yields consistent performance gains that justify their additional computational and operational overhead in churn prediction settings. Third, we plan to investigate more expressive hybrid and ensemble architectures, combining representation learning with gradient-boosting or attention-based mechanisms, while maintaining interpretability and calibration constraints required for deployment. These extensions will be complemented by deeper analysis of cross-dataset calibration stability, segment-level performance consistency, and cost-sensitive optimization to further align predictive modeling with business decision-making.

5 Conclusion

We set out to answer four questions and the evidence supports clear conclusions. For RQ1 which tabular model performs best under a rigorous evaluation protocol a compact multilayer perceptron consistently led the benchmark while strong tree ensembles followed, confirming that modest neural capacity can capture nonlinear structure in telecom churn without sacrificing stability. For RQ2 whether gains are statistically reliable fold wise Wilcoxon tests embedded in the nested cross validation workflow indicate consistent advantages for the multilayer perceptron with two sided results at trend level and one sided tests in the expected direction, supporting practical significance rather than a favourable split. For RQ3 which factors drive churn and how they interact SHAP together with partial dependence and individual conditional expectation shows a monotonic risk decrease with tenure, a sharp rise from zero to five technical tickets that then saturates, and amplifying effects from high charges, month to month contracts, and fiber access, yielding coherent global and local explanations. For RQ4 how insights translate

into interventions we map these patterns to policy by prioritizing contract migration for high charge new customers, proactive ticket triage once a customer crosses two to five tickets, targeted credits for early tenure fiber subscribers, and risk value sequencing of outreach to focus limited capacity where expected return is highest. The contribution is a reproducible pipeline that couples strong tabular learners with model agnostic explanations that are auditor friendly and operationally actionable. Future work will extend this foundation with cost sensitive learning and explicit thresholding tied to treatment budgets, uplift modeling to target customers who are both at risk and persuadable, causal what if analysis to test policy scenarios before rollout, survival style time to churn modeling for timing of offers, and streaming retraining with drift detection and fairness auditing to keep models reliable over time.

Author contributions

Ismoth Zerine contributed to Validation, Visualization, Resources, and Writing—review & editing by reproducing key experiments, developing explainability analyses (SHAP, PDP/ICE), conducting error analysis (FP/FN patterns and class-imbalance effects), and refining plots and reproducibility documentation; Md Mainul Islam contributed to Data curation, Software, Investigation, Validation, and Writing—review & editing by preparing the Kaggle Telco dataset and preprocessing pipeline, implementing and tuning baseline learners (Logistic Regression with Elastic Net, RBF-SVC, Random Forest, XGBoost), and conducting robustness checks and ablations across folds; Mohammad Aman Ullah Khan contributed to Conceptualization, Methodology, Formal analysis, Software, Visualization, and Writing—original draft by leading the modeling and evaluation framework, executing nested cross-validation, implementing the multilayer perceptron and Wilcoxon signed-rank testing, and preparing the primary figures and tables; Md Asikur Rahman Chy, Abu Saleh Muhammad Saimon, and Mia Md. Tofayel Gonee Manik contributed during the revision stage to Writing—review & editing by improving manuscript clarity, strengthening technical explanations, and assisting with formatting and consistency corrections in response to reviewer feedback; Chala Wata contributed to Supervision, Project administration, Resources, and Writing—review & editing by supervising study design and interpretation, aligning findings with operational telecom use-cases and actionable recommendations, coordinating author contributions, and serving as corresponding author. All authors approved the final manuscript and agree to be accountable for all aspects of the work.

Funding

This research received no external funding.

Data availability

The dataset used in this study is publicly available at <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>.

Declarations

Ethics approval

This study does not involve human participants, human biological material, or personal identifiable information. The research is based exclusively on a secondary analysis of a publicly available, anonymized dataset. Therefore, ethical approval was not required.

Consent to participate

Not applicable. The study uses fully anonymized secondary data and does not involve direct interaction with human participants.

Consent to publication

Not applicable. The manuscript does not contain any individual-level data, images, or information that could identify participants.

Competing interests

The authors declare no competing interests.

Received: 2 October 2025 / Accepted: 4 February 2026

Published online: 22 February 2026

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